

Research Readiness Statement

a. Degree program and mentor preparation. Colton Goodrich is a PhD student in Geosensing Systems Engineering & Sciences at the University of Houston, advised by the proposal PI, Craig L. Glennie. Graduate coursework in remote sensing and geosensing, together with prior degrees in Geology (M.S. and B.S., Brigham Young University) and a Geospatial Analysis certificate (Weber State University), provide a foundational understanding of terrain analysis, digital elevation models, and remote sensing relevant to the proposed research. The proposed project requires use of the Python scientific stack (NumPy, GDAL, Rasterio, PyTorch), PDAL, WhiteboxTools, and QGIS, together with DEM differencing, co-registration (ICP), and robust statistics; the FI is already proficient in each of these tools and methods. Two areas require further preparation — hierarchical/Bayesian statistical modeling and cold-regions (Antarctic) process geomorphology; the FI will address these deficiencies through graduate coursework and directed study with the PI.

b. Graduate study timeline. Degree type: PhD. Subject area: Geosensing Systems Engineering & Sciences. Enrollment: enrolled since August 2025 (approximately one year at the proposal due date). Estimated graduation: August 2029.

c. Other relevant experience. The FI independently designed and built an end-to-end airborne-lidar terrain-analysis pipeline on the Appalachian Plateau — point-cloud processing (PDAL), bare-earth derivative generation (GDAL/WhiteboxTools), hand-labeled training-set construction (QGIS), U-Net and instance-segmentation model training (PyTorch), and DEM-of-Difference change analysis with ICP co-registration and NMAD-based uncertainty — an independent research project that exercises the methods used in the proposed work. Additional relevant experience includes InSAR-based DEM generation and accuracy assessment against an airborne lidar reference; five years as an Environmental Scientist at the Utah Division of Oil, Gas & Mining (geospatial data analysis and Python automation); prior work as an Earth Scientist at Chevron; a graduate teaching assistantship at Brigham Young University; and a Python Programming course (The Tech Academy, 2024).

NASA BIOGRAPHICAL SKETCH FORM

Identifying Information

Name: Goodrich, Colton, Lynn

Persistent Identifier (PID): ORCID 0009-0006-5237-2836 (<https://orcid.org/0009-0006-5237-2836>)

Position Title: Doctoral Researcher, Geosensing Systems Engineering & Sciences

Organization and Location: University of Houston, Houston, TX, USA

Professional Preparation

University of Houston — Houston, TX, USA

Degree: Ph.D., Geosensing Systems Engineering & Sciences. Start: 08/2025. Received: Expected 08/2029. Field: Geosensing / Remote Sensing

Weber State University — Ogden, UT, USA

Degree: Certificate, Geospatial Analysis. Start: 2018. Received: 2019. Field: Geospatial Analysis / GIS

Brigham Young University — Provo, UT, USA

Degree: M.S. Start: 09/2011. Received: 12/2013. Field: Geology

Brigham Young University — Provo, UT, USA

Degree: B.S. Start: 2006. Received: 2011. Field: Geology

Appointments and Positions

2025–Present · Doctoral Researcher

University of Houston, Geosensing Systems Engineering & Sciences · Houston, TX, USA

2020–2025 · Environmental Scientist III (promoted from Environmental Scientist II)

Utah Department of Natural Resources, Division of Oil, Gas & Mining · Syracuse, UT, USA

2013–2016 · Earth Scientist

Chevron Energy Company · Houston, TX, USA

2012–2012 · Reservoir Modeling Intern

Chevron Energy Company · Houston, TX, USA

2011–2013 · Graduate Teaching Assistant

Brigham Young University, Department of Geology · Provo, UT, USA

Products

1. Goodrich, C. (2025). WellSight: An autonomous lidar analysis pipeline for detecting orphaned and abandoned oil & gas wells [software]. Multi-task deep-learning system (U-Net, ~8M parameters) segmenting well pits, access roads, and well pads from lidar-derived rasters at 0.5–1 m resolution; integrates PDAL, WhiteboxTools, GDAL/Rasterio, and PyTorch.
https://github.com/clgoodrich/lidar_project

2. Goodrich, C. (2025). SAOCOM L-band InSAR digital elevation model generation and validation framework [software/dataset]. P-SBAS time-series interferometry of SAOCOM-1A/1B data producing high-coherence, vegetation-penetrating DEMs; accuracy assessment (NMAD, RMSE, CORINE land-cover-stratified) against TINITALY lidar and Copernicus GLO-30, in support of the NASA Surface Topography and Vegetation (STV) Decadal Survey. https://github.com/clgoodrich/saocom_project
3. Goodrich, C. (2025). Coastal land-subsidence mapping from Small Baseline Subset (SBAS) InSAR time series [software]. Deformation-rate estimation on HyP3 SAR processing outputs with an accuracy-validation workflow. https://github.com/clgoodrich/Final_project_coastal
4. Goodrich, C. (2024). EToolsV3 — Regulatory GIS web application for directional-survey trajectory computation and Well Completion Report generation [software]. Minimum-curvature trajectory recomputation, PLSS location, boundary-footage calculation, LLM-assisted PDF extraction, interactive 2D/3D plats (FastAPI, NiceGUI, Shapely/GeoPandas). <https://github.com/clgoodrich/EToolsV3>
5. Goodrich, C. (2024). ArcGIS geoprocessing toolbox for social-media geospatial analysis [software]. Parameterized ArcPy tools (kernel-density hotspots, H3 hexagon aggregation, time-enabled mosaics). https://github.com/clgoodrich/tweet_project_v2
6. Goodrich, C. (2024). 1 m DEM/hillshade mosaic over Oil Creek State Park, Pennsylvania [dataset]. Contiguous 22-tile (~7.5 × 9 km) mosaic assembled via a greedy contiguous-patch downloader over the USGS TNM Access API. https://github.com/clgoodrich/lidar_project
7. Goodrich, C. (2024). OGM Board regulatory mapping and visualization application [software]. Decision-support application presenting well data, analyses, and digital maps for monthly hearings of the Utah Board of Oil, Gas & Mining. <https://github.com/clgoodrich/OGMBoardProject>

Certification

I certify that the information provided is current, accurate, and complete. This includes but is not limited to information related to domestic and foreign appointments and positions.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

Signature: Colton Goodrich Date: 07/14/2026

*Privacy Act and Burden Statement:

The information requested on this form is solicited under the authority of the National Aeronautics and Space Act of 1958 (51 U.S.C. 20101 et. seq.) The information on this form will be used in connection with the evaluation of qualified proposals. The information requested may be disclosed to qualified reviewers for their opinion and evaluation of applicants and their proposals as part of the NASA application review process; to other government agencies or other entities needing information regarding applicants or nominees as part of a joint application review process, or in order to coordinate programs or policy; or to individual or institutional applicants and award recipient institutions to provide or obtain data as part of the application review process, award decisions, or administering awards. Additionally, information requested may be disclosed to other entities when merging records with other computer files to carry out studies for or otherwise assist NASA with program management, evaluation, or reporting as well as contractors,

grantees, volunteers, experts, consultants, advisors, and other individuals who perform a service to or work on or under a contract, grant, cooperative agreement, advisory committee, independent review boards, or other arrangement with or for NASA or for the Federal government. See NASA Systems of Records Notice (SORN) NASA 100AAR, "Opportunities and Associated Reviewers."

This information collection meets the requirements of 44 U.S.C. § 3507 as amended by section 2 of the Paperwork Reduction Act of 1995. You do not need to answer these questions unless we display a valid Office of Management and Budget (OMB) control number. The OMB control number for this collection is 3145-0279 and this information collection expires on October 31, 2026. We estimate that it will take two (2) hours to read the instructions, gather the facts, and answer the questions. Send only comments relating to our time estimate to: christiane.diallo@nasa.gov.

IDENTIFYING INFORMATION:

NAME: Glennie, Craig

PERSISTENT IDENTIFIER (PID): <https://orcid.org/0000-0003-1570-0889>

POSITION TITLE: Professor

PRIMARY ORGANIZATION AND LOCATION: The University of Houston, Civil & Environmental Engineering, Houston, Texas, United States**Professional Preparation:**

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
The University of Calgary, Calgary, Alberta, AB, Canada	PHD	06/1996 - 06/1999	Geomatics Engineering
The University of Calgary, Calgary, Alberta, AB, Canada	B.Sc. Engineering	09/1992 - 06/1996	Geomatics Engineering

Appointments and Positions

2020 - present Professor, The University of Houston, Civil & Environmental Engineering, Houston, Texas, United States

2024 - present Hugh Roy and Lillie Cranz Cullen Distinguished University Chair, University of Houston, Houston, TX, United States

2016 - 2020 Associate Professor, The University of Houston, Houston, TX, United States

2010 - 2016 Assistant Professor, The University of Houston, Houston, TX, United States

2007 - 2010 Vice President of Engineering, Terrapoint, Spring, TX, United States

2004 - 2007 Director of Operations, Terrapoint, Spring, TX, United States

2000 - 2004 Senior Geomatics Engineering, Aerotec LLC, Birmingham, Alabama, United States

1999 - 2000 Geomatics Engineer, Focus Surveys, Calgary, Alberta, AB, Canada

Products

1. Okyay U, Telling J, Glennie C, Dietrich W. Airborne lidar change detection: An overview of Earth sciences applications. Earth-Science Reviews. 2019 November 01; 198:102929. Available from: <https://www.sciencedirect.com/science/article/pii/S0012825218306470> issn: 0012-8252
2. Levy J, Fountain A, Obryk M, Telling J, Glennie C, Pettersson R, Gooseff M, Van Horn D. Decadal topographic change in the McMurdo Dry Valleys of Antarctica: Thermokarst subsidence, glacier thinning, and transfer of water storage from the cryosphere to the hydrosphere. Geomorphology. 2018 December 15; 323:80-97. Available from: <https://www.sciencedirect.com/science/article/pii/S0169555X18303696> issn: 0169-555X
3. Bui L, Glennie C. Estimation of lidar-based gridded DEM uncertainty with varying terrain roughness and point density. ISPRS Open Journal of Photogrammetry and Remote Sensing. 2023 January 01; 7:100028. Available from: <https://www.sciencedirect.com/science/article/pii/S2667393222000175> issn: 2667-3932
4. Glennie Craig. Arctic High-Resolution Elevation Models: Accuracy in Sloped and Vegetated

Terrain. Journal of Surveying Engineering. 2018 February 01; 144(1):06017003. Available from: [https://doi.org/10.1061/\(ASCE\)SU.1943-5428.0000245](https://doi.org/10.1061/(ASCE)SU.1943-5428.0000245) DOI: 10.1061/(ASCE)SU.1943-5428.0000245

5. Barlow M, Zhu X, Glennie C. Stream Boundary Detection of a Hyper-Arid, Polar Region Using a U-Net Architecture: Taylor Valley, Antarctica. Remote Sensing. 2022 January 05; 14(1):234-. Available from: <https://www.mdpi.com/2072-4292/14/1/234> DOI: 10.3390/rs14010234
6. Telling J, Lyda A, Hartzell P, Glennie C. Review of Earth science research using terrestrial laser scanning. Earth-Science Reviews. 2017 June 01; 169:35-68. Available from: <https://www.sciencedirect.com/science/article/pii/S0012825216304433> issn: 0012-8252
7. Butler H, Chambers B, Hartzell P, Glennie C. PDAL: An open source library for the processing and analysis of point clouds. Computers & Geosciences. 2021 March 01; 148:104680. Available from: <https://www.sciencedirect.com/science/article/pii/S0098300420306518> issn: 0098-3004
8. N. Ekhtari, C. Glennie, J. C. Fernandez-Diaz. Classification of Airborne Multispectral Lidar Point Clouds for Land Cover Mapping. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing. 2018; 11(6):2068-2078. DOI: 10.1109/JSTARS.2018.2835483
9. Glennie C, Hinojosa-Corona A, Nissen E, Kusari A, Oskin M, Arrowsmith J, Borsa A. Optimization of legacy lidar data sets for measuring near-field earthquake displacements. Geophysical Research Letters. 2014 May 29; 41(10):3494-3501. Available from: <https://agupubs.onlinelibrary.wiley.com/doi/10.1002/2014GL059919> DOI: 10.1002/2014GL059919
10. Haces-Garcia F, Glennie C, Rifai H, Hoskere V. The longitudinal assessment of flood hazard in cities: Unlocking the floodplain record of Houston, TX, USA. Journal of Hydrology: Regional Studies. 2026 April; 64:103113-. Available from: <https://linkinghub.elsevier.com/retrieve/pii/S221458182600011X> DOI: 10.1016/j.ejrh.2026.103113

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

Certified by Glennie, Craig in SciENcv on 2026-07-14 12:36:55

NASA CURRENT AND PENDING (OTHER) SUPPORT FORM

Identifying Information

Name: Goodrich, Colton, Lynn

Persistent Identifier (PID): ORCID 0009-0006-5237-2836

Position Title: Doctoral Candidate, Geosensing Systems Engineering & Sciences

Organization and Location: University of Houston, Houston, TX, USA

a. Proposals and Active Projects

Entry 1 — Graduate Assistantship

Title: Graduate assistantship — monthly stipend and tuition waiver

Status of Support: Current

Award Number (if available):

Source of Support: University of Houston, Cullen College of Engineering, NCALM

Primary Place of Performance: University of Houston, Houston, TX, USA

Start Date: 08/2025 **Expected End Date:** 08/2029

Total Anticipated Amount: Stipend \$2,601.67/month (~\$31,220/year, 12-month appointment) + tuition waiver ~\$3,852/semester (~\$7,700/year; TX-resident Engineering-PhD rate, 9 SCH, AY2027; mandatory fees additional)

Person-Months Per Year: 12

Overall Objectives: Graduate assistantship supporting the FI's doctoral study, training, and dissertation research in geosensing/remote sensing; not tied to a single external project.

Statement of Potential Overlap: No scientific overlap with the pending FINESST proposal (Entry 2). If FINESST is awarded, 6 person-months/year now covered by this assistantship would be reallocated to FINESST for the proposed research; the assistantship adjusts accordingly, so the FI's total committed effort stays at 12 person-months/year with no double compensation.

Entry 2 — This FINESST proposal

Title: Detection to Attribution: Linking Two Decades of Antarctic Stream-Channel Change to Climate Drivers Utilizing Terrain Data

Status of Support: Pending

Award Number (if available): N/A (under review)

Source of Support: National Aeronautics and Space Administration (NASA) — FINESST, ROSES-2025 Appendix F.5

Primary Place of Performance: University of Houston, Houston, TX, USA

Start Date: 01/2027 **End Date:** 12/2029 (3 years)

Total Anticipated Amount: up to \$148,368 (\$48,456 in year 1 and \$49,956 in years 2 and 3, under the \$50,000 annual cap)

Person-Months Per Year: 6

Overall Objectives: Attribute two decades of measured stream-channel and stream-corridor geomorphic change in the McMurdo Dry Valleys to climate drivers, using multi-epoch lidar and satellite-DEM differencing with per-pixel calibrated uncertainty.

Statement of Potential Overlap: This is the proposal under consideration. No scientific overlap with current support; any budgetary/effort overlap with the assistantship is resolved by reallocation so total committed effort remains ≤ 12 person-months/year.

Certification

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Signature: Colton Goodrich Date: 07/14/2026

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CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Glennie, Craig

PERSISTENT IDENTIFIER (PID): <https://orcid.org/0000-0003-1570-0889>

*POSITION TITLE: Professor

*ORGANIZATION AND LOCATION: The University of Houston, Houston, Texas, United States

Proposals and Active Projects

*Proposal/Active Project Title: STV Incubation Study Team Technology Co-Lead
2025-2029

*Status of Support: Current

Proposal/Award Number:

*Source of Support: NASA

*Primary Place of Performance: Houston, TX

*Proposal/Active Project Start Date: (MM/YYYY): 10/2025

*Proposal/Active Project End Date: (MM/YYYY): 09/2029

*Total Anticipated Proposal/Project Amount: \$227,479

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1
2027	1
2028	1
2029	1

***Overall Objectives:**

Technology Co-Lead for Decadal Survey Incubation Study for development of a high-resolution satellite based 3D observing system for measuring surface topography and vegetation (STV).

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:** Enhancing Graduate Geodesy at The University of Houston

***Status of Support:** Current

Proposal/Award Number:

***Source of Support:** NGA

***Primary Place of Performance:** University of Houston

***Proposal/Active Project Start Date: (MM/YYYY):** 06/2025

***Proposal/Active Project End Date: (MM/YYYY):** 05/2030

***Total Anticipated Proposal/Project Amount:** \$2,132,116

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	0.5
2026	0.5
2027	0.5
2028	0.5
2029	0.5

***Overall Objectives:**

The objective of this proposal is to establish a graduate research student pipeline for the MSc and PhD programs in Geosensing Systems Engineering and Sciences (GSES) at the Cullen College of Engineering (CCE) at the University of Houston (UH); focusing on geodetic practice through coursework, research and outreach activities.

***Statement of Potential Overlap:**

Students in this current project, if it continues, will be working in tandem with the proposed students. These synergies will allow more momentum for a graduate geodesy pipeline of professionals.

***Proposal/Active Project Title:** Facility Support to Renew Operation of the National Center for Airborne Laser Mapping (NCALM)

***Status of Support:** Current

Proposal/Award Number: 2324629

***Source of Support:** NSF

***Primary Place of Performance:** University of Houston

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2024

***Proposal/Active Project End Date: (MM/YYYY):** 08/2027

***Total Anticipated Proposal/Project Amount:** \$1,062,651

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	1
2026	1
2027	0.1

***Overall Objectives:**

Since NCALM inception in 2003 it has had three objectives: 1) To train and educate graduate students to meet the needs of private industry, government agencies, and academic institutions. 2) To advance the state of the art in airborne lidar technology and its application by the research community, and 3) To provide research quality airborne lidar data to the scientific community.

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:** Analysis Capabilities for Competition, Crisis, and Combat (AC4) (PI)

***Status of Support:** Current

Proposal/Award Number: W911QX23D0009

***Source of Support:** Army Devcom Analysis Center

***Primary Place of Performance:** University of Houston

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2023

***Proposal/Active Project End Date: (MM/YYYY):** 09/2028

***Total Anticipated Proposal/Project Amount:** \$63,500,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2023	2
2024	1
2025	1

Year	Person Months
2026	1
2027	1

***Overall Objectives:**

Modeling and Simulation for Devcom Analysis Center. Providing frameworks and assessment schemes for Army needs during competition and crisis.

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:** Collaborative Research: Geomorphological Change in in the McMurdo Dry Valleys: AI-Based Stream Mapping and Multi-Temporal Topography to investigate fluvial thermokarst changes

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** NSF

***Primary Place of Performance:** University of Houston

***Proposal/Active Project Start Date: (MM/YYYY):** 01/2027

***Proposal/Active Project End Date: (MM/YYYY):** 12/2029

***Total Anticipated Proposal/Project Amount:** \$325,303

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	0.75
2028	0.75
2029	0.75

***Overall Objectives:**

The McMurdo Dry Valleys (MDV) are the largest ice-free region in Antarctica and are increasingly recognized as a landscape on the threshold of climate-driven change which could lead to reorganizing of biogeochemical systems. We have developed a new, scalable framework for detecting stream channel change by combining deep-learning stream-boundary segmentation, co-registration of multi-source DEMs, and uncertainty-aware topographic differencing. Using lidar-based digital elevation models we predict the location and extent of streams in the MDV. From multi-year photogrammetric digital elevation models, the elevation change within the channels are calculated

and net rates of stream incision and deposition calculated. We hypothesize that the largest geomorphic changes occur where meltwater pathways intersect ice-cemented or ice-rich permafrost in the Coastal Thaw Zone, whereas drier terrain in inland valleys will show less sediment mobility and weaker erosion. The project will produce valley-wide maps of geomorphic sensitivity to fluvial erosion and predictive models for where incision, bank failure, aggradation, widening, and lateral migration are most likely to accelerate under continued warming. This work relies on existing data and no field work is requested.

***Statement of Potential Overlap:**

This proposed project.

***Proposal/Active Project Title:** Integrated Multi-Sensor Monitoring of Arctic Coastal Hazards

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** ESTCP

***Primary Place of Performance:** Houston, TX

***Proposal/Active Project Start Date: (MM/YYYY):** 01/2027

***Proposal/Active Project End Date: (MM/YYYY):** 12/2028

***Total Anticipated Proposal/Project Amount:** \$1,951,616

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	0.5
2028	0.5

***Overall Objectives:**

Establish a baseline coastal monitoring framework for the Utqiagvik and Prudhoe Bay coastlines by integrating multiple sensors, including airborne lidar, satellite imagery, and prior coastal surveys, to quantify shoreline position and coastal morphology.

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:** Advancing Geodesy Workforce Development Through an Education-to-Careers Pipeline at the University of Houston

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** NGA

***Primary Place of Performance:** University of Houston

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2026

***Proposal/Active Project End Date: (MM/YYYY):** 08/2031

***Total Anticipated Proposal/Project Amount:** \$2,429,129

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	0.5
2028	0.5
2029	0.5
2030	0.5
2031	0.5

***Overall Objectives:**

This project will establish a sustainable workforce pipeline of students trained in mission-relevant geodesy applications. We will provide structured entry-level research experiences and advanced graduate education to develop professionals uniquely qualified to support NGA's workforce needs.

***Statement of Potential Overlap:**

None

***Proposal/Active Project Title:** Gulf Artificial Intelligence Learning Center for Severe Weather and Infrastructure Systems (GAIL-CSWIS)

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** DOE

***Primary Place of Performance:** University of Houston

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2026

***Proposal/Active Project End Date: (MM/YYYY):** 08/2029

***Total Anticipated Proposal/Project Amount:** \$5,000,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	1
2028	1
2029	1

***Overall Objectives:**

This proposal responds to DOE's call for proposals and establishes the Gulf Artificial Intelligence Learning Center for Severe Weather and Infrastructure Systems (GAIL-CSWIS). CSWIS will serve as a multidisciplinary and innovation hub dedicated to advancing the understanding, prediction, and mitigation of severe weather-related hazards that threaten energy, power, and water systems, and other critical infrastructure across the Gulf Region.

***Statement of Potential Overlap:**

This proposal

***Proposal/Active Project Title:** Creating a NASA Surface Topography and Vegetation (STV) Novel Observing System (NOS)

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** NASA

***Primary Place of Performance:** Houston, TX

***Proposal/Active Project Start Date: (MM/YYYY):** 06/2026

***Proposal/Active Project End Date: (MM/YYYY):** 05/2029

***Total Anticipated Proposal/Project Amount:** \$4,206,117

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.25
2027	0.25
2028	0.25

***Overall Objectives:**

To develop a novel observing system (NOS) demonstration to enable the implementation of NASA's Surface Topography and Vegetation (STV) Targeted Observable.

***Statement of Potential Overlap:**

None

Certification:

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Certified by Glennie, Craig in SciENcv on 2026-07-14 12:35:57

FINESST Ancillary Documents

A. Facilities, Equipment, and Resources Statement

The proposed work requires modest, presently available resources. There is no need for facility development or shared instrument time. The FI has easy access to all of the resources listed below, so no resource-support letter is required.

Computing. The FI has a workstation with a single consumer-grade GPU (NVIDIA GeForce GTX 1070 Ti, 8 GB of VRAM) and at least 2 TB of local storage, owned by the research group and the FI. This is more than sufficient for all model training and elevation processing in the proposal.

Workspace. The FI has office and lab space in NCALM (National Center for Airborne Laser Mapping) at the University of Houston, with institutional backup storage for project data.

Software. The full processing stack is open source and requires no subscriptions or commercial licenses (Python scientific stack, PDAL, GDAL/OGR, WhiteboxTools, PyTorch, QGIS).

Field work. None is proposed. The project is based entirely on existing datasets and requires no Antarctic deployment, logistics support, or permitting.

B. Acknowledgements

The Future Investigator (FI) is the primary author of this proposal. The FI thanks Dr. Craig Glennie for his scientific mentorship and guidance in shaping and editing the research, and Cami Barlow, whose dissertation on Antarctic stream-channel detection and geomorphic change analysis provides the foundation this project builds upon. AI assistance (Anthropic's Claude, used as an editing tool under the FI's direction) contributed to light grammar and structure editing, document formatting, and data-availability verification; all scientific content, claims, and decisions are the FI's own and were reviewed by the FI and PI. Publicly archived datasets are credited to their providers: NASA ATM (via USGS), NCALM/OpenTopography, the Polar Geospatial Center (REMA), the McMurdo Dry Valleys LTER (via the Environmental Data Initiative), Copernicus/ECMWF (ERA5), and NCAR (AMPS).

Budget Justification

The University of Houston's definition of a calendar year is the 12-month period starting on January 1 and ending on December 31. The University of Houston's definition of a fiscal year is the 12-month period starting on September 1 and ending on August 31. A University Year, as used in connection with faculty and other personnel appointments, contains any two terms in the calendar for the year in question.

Personnel. Colton Goodrich, the Future Investigator (FI), will carry out his work under the supervision of the PI. The FI requests 12-months funding in each of the three years to support his research. The funding will be used to cover the FI's monthly stipend and fringe benefits.

The salaries and benefits for the FI are determined based on the institution's current salary and fringe scales.

Domestic Travel. We request an annual allocation of \$2,500 for the FI to attend a national meeting, such as the American Geophysical Union or the Geological Society of America annual meeting. Meeting attendance provides valuable opportunities for the FI to be exposed to the community and interact with other scientists. Cost estimates are based on prior experience.

The costs for the FI to the meeting are estimated as follows: Airfare - \$500; Ground transportation - \$175; Hotel - \$200/night ×5 nights; Registration - \$450; Per Diem - \$75/day ×5 days. The total is \$2,500, or \$7,500 over the three-year project.

Materials and Supplies. We request \$500 in each year of the project to support miscellaneous costs associated with this project, including photocopy charges, thumb drives, research-related supplies, etc.

Publication Costs. We request \$1,500 in each of years 2 and 3 of the project to support the cost of publishing articles in peer-reviewed journals.

Tuition and Fees. We request \$8,736 in each year of the project to cover tuition and fees for 9 semester hours during each fall and spring semester.

Indirect Costs. None.

Cost Category. The University of Houston treats the FI as a participant. All FI costs are requested as participant support costs as defined in 2 CFR 200.1, entered on the NSPIRES cover page under E. Direct Costs-Participant/Trainee Support Cost. No indirect costs are requested.

Budget Summary

Category	Year 1	Year 2	Year 3	Total
FI stipend and fringe	\$36,720	\$36,720	\$36,720	\$110,160
Domestic travel	\$2,500	\$2,500	\$2,500	\$7,500

Tuition and fees	\$8,736	\$8,736	\$8,736	\$26,208
Materials and supplies	\$500	\$500	\$500	\$1,500
Publication costs	\$0	\$1,500	\$1,500	\$3,000
Indirect costs	\$0	\$0	\$0	\$0
Total	\$48,456	\$49,956	\$49,956	\$148,368